

Unit-4



**Ganpat
University**

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Department of
Computer Science



B.Sc. Hons. (CA&IT) Sem-II
Marginal Costing
UNIT-4

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Marginal Costing

- Marginal Cost is defined as, ‘ the change in aggregate costs due to change in the volume of production by one unit.’
- **Marginal cost is the additional cost incurred in the production of one more unit of a good or service.**
- It is derived from the variable cost of production, given that fixed costs do not change as output changes, hence no additional fixed cost is incurred in producing another unit of a good or service once production has already started.

Example

- Cost to produce 10 cups of coffee = ₹100
- Cost to produce 11th cup = ₹8
- Marginal Cost = ₹8 (cost of one additional unit)

Main Features of Marginal Costing

- Only variable costs are considered for product costing.
- Fixed costs remain unchanged regardless of output level.
- Profit is calculated using Contribution.
- Useful for short-term decision making.

Fixed Cost

Fixed costs are costs that remain constant irrespective of production level.

Characteristics

- Do not change with increase or decrease in output
- Incurred for a specific period
- Controlled by management decisions

Examples

- Rent
- Salaries
- Insurance
- Interest on loan
- Lease payments

Variable Cost

Variable costs are costs that change directly and proportionately with output.

Characteristics

- Increase with increase in production
- Decrease with decrease in production
- Directly related to activity level

Examples

- Direct materials
- Direct labour
- Raw materials
- Commission
- Power and fuel

Contribution

- Contribution is the difference between sales and variable cost.
- Contribution is also known as “Contribution Margin” or “Gross Margin”.
- $\text{Contribution} = \text{Sales} - \text{Variable Cost}$
- Contribution is the profit before adjusting the Fixed Costs.
- If the Fixed Cost is equal to contribution than such situation is known as Break- even.

Profit-volume ratio

- Profit-volume ratio indicates the relationship between contribution and sales and is usually expressed in percentage.
- High P/V ratio indicate high profitability
- Low P/V ratio indicate low profitability

When would P/V Ratio change?

It changes only if:

- Selling price changes
- Variable cost per unit changes
- **If Selling Price and Variable Cost per unit remain constant → P/V Ratio will always remain constant**, no matter how many units are sold.

BREAK EVEN POINT

- It is the volume of output at which neither a profit nor a loss is earned.
- It means contribution covers the fixed cost.

- At Break Even Point, Contribution = Fixed Cost

Break Even Point may be presented in two ways :

- The number of units that should be sold so that incomes and expenses are equal.
- The amount of sales value in rupees so that incomes and expenses are equal.

Margin of Safety

- Margin of Safety (MOS) measures the distance between budgeted sales and breakeven sales
- The difference between **Actual Sales and Break-Even Sales**.
- It shows **how much sales can decrease before the business reaches the break-even point**.

Metric	Formula	400 Units (Loss)	500 Units (BEP)	800 Units (Profit)
Sales Value	$\text{Units} \times \$20$	\$8,000	\$10,000	\$16,000
Variable Cost	$\text{Units} \times \$12$	\$4,800	\$6,000	\$9,600
Contribution (C)	$\text{Sales} - \text{Variable Cost}$	\$3,200	\$4,000	\$6,400
Fixed Cost (FC)	Actual Amount	\$4,000	\$4,000	\$4,000
Profit / (Loss)	$C - FC$	(\$800)	\$0	\$2,400
P/V Ratio	$(C \div \text{Sales}) \times 100$	40%	40%	40%
BEP (Value)	$FC \div \text{P/V Ratio}$	\$10,000	\$10,000	\$10,000
Margin of Safety	$\text{Sales} - \text{BEP Sales}$	(\$2,000)	\$0	\$6,000

Advantages of Marginal Costing

1) Fixed Cost Remains Unchanged

- Fixed cost does not increase or decrease with production volume.

2) Helpful in Profit Planning (Prosperity Period)

- Helps management choose orders giving maximum contribution.

3) Useful During Depression

- Shows the minimum price at which goods can be sold without loss.

4) Make or Buy Decision

- Helps decide whether to manufacture internally or buy from outside.

5) Dumping in Foreign Market

- Helps compare foreign market price with marginal cost.

6) Relationship Between Volume and Profit

- Clearly shows how change in sales volume affects profit.

7) Cost Control

- Cost divided into fixed and variable helps easy control.

8) Easy Pricing Decisions

- Sales manager can easily fix selling price using contribution.

Limitations of Marginal Costing

- Difficult to clearly classify costs into fixed and variable.
- Uses historical data, while decisions relate to future.
- Not suitable for long-term pricing decisions.
- Ignores fixed cost in product valuation.
- Possibility of loss if used blindly.
- No standard method for performance evaluation.

Formulas

Concept	Meaning / Definition	Formula
Contribution	Difference between Sales and Variable Cost	$\text{Contribution} = \text{Sales} - \text{Variable Cost}$ $\text{Total contribution} = \text{Total units} * \text{contribution per unit}$
P/V Ratio (Profit-Volume Ratio)	Shows profitability of sales To know how profit changes with change in sales	$\text{P/V Ratio} = (\text{Contribution} / \text{Sales}) \times 100$

Break-Even Point (BEP) - Units	Minimum units to sell to avoid loss no profit and no loss	$BEP \text{ (Units)} = \text{Fixed Cost} / \text{Contribution per Unit}$
Break-Even Point (BEP) - Value	Sales value where no profit and no loss	$BEP \text{ (₹)} = \text{Fixed Cost} / \text{P/V Ratio}$
Margin of Safety (MoS)	-Excess of actual sales over break-even sales -Shows how much sales can fall before loss starts.	$MoS = \text{Actual Sales} - \text{BEP Sales}$
Profit	Excess of contribution over fixed cost	$\text{Profit} = \text{Contribution} - \text{Fixed Cost}$

Example

- Variable cost per unit = ₹200
- Selling price per unit = ₹500
- Fixed cost = ₹30,000
- Units sold = 200 units

(a) Contribution per Unit
 $= 500 - 200 = ₹300$

(b) Total Contribution
 $= 200 \times 300 = ₹60,000$

(c) P/V Ratio
 $= (300 / 500) \times 100 = 60\%$

(d) Profit
 $= 60,000 - 30,000 = ₹30,000$

(e) Break Even Point (Units)
 $= 30,000 / 300 = 100 \text{ units}$

(f) Break Even Point (₹)

$$= (30,000 \times 100) / 60 = ₹50,000$$

(g) Margin of Safety

$$= \text{Actual Sales} - \text{BEP Sales} = (500 \times 200) - 50,000 = 1,00,000 - 50,000 = ₹50,000$$

Examples

1) A company produces a single product.

The following information is given:

- Selling Price per unit = ₹600
- Variable Cost per unit = ₹350
- Fixed Cost = ₹50,000
- Actual Sales = 400 units

Calculate:

1. Contribution per unit
2. Total Contribution
3. P/V Ratio
4. Break Even Point (Units)
5. Break Even Point (₹)
6. Margin of Safety (₹)
7. Profit or Loss

1 Contribution per Unit

Formula:

Contribution per unit = Selling Price – Variable Cost

$$= 600 - 350 = \boxed{\text{₹}250}$$

2 Total Contribution

Formula:

Total Contribution = Units Sold × Contribution per unit

$$= 400 \times 250 = \boxed{\text{₹}1,00,000}$$

3 P/V Ratio

Formula:

$$\begin{aligned}\text{P/V Ratio} &= \frac{\text{Contribution per unit}}{\text{Selling Price}} \times 100 \\ &= \frac{250}{600} \times 100 = \boxed{41.67\%}\end{aligned}$$

4 Break Even Point (Units)

Formula:

$$\begin{aligned}\text{BEP (Units)} &= \frac{\text{Fixed Cost}}{\text{Contribution per unit}} \\ &= \frac{50,000}{250} = \boxed{200 \text{ units}}\end{aligned}$$

5 Break Even Point (₹)

Formula:

$$\begin{aligned}\text{BEP (₹)} &= \frac{\text{Fixed Cost}}{\text{P/V Ratio}} \\ &= \frac{50,000}{41.67\%} = \boxed{\text{₹}1,20,000 \text{ (approx)}}\end{aligned}$$

6 Margin of Safety (₹)

Formula:

$$\text{MoS} = \text{Actual Sales} - \text{BEP Sales}$$

$$\text{Actual Sales} = 400 \times 600 = \text{₹}2,40,000$$

$$= 2,40,000 - 1,20,000 = \boxed{\text{₹}1,20,000}$$

7 Profit or Loss

Formula:

$$\begin{aligned}\text{Profit} &= \text{Contribution} - \text{Fixed Cost} \\ &= 1,00,000 - 50,000 = \boxed{\text{₹}50,000 \text{ (Profit)}}\end{aligned}$$

Particulars	Answer
Contribution per unit	₹250
Total Contribution	₹1,00,000
P/V Ratio	41.67%
BEP (Units)	200 units
BEP (₹)	₹1,20,000
Margin of Safety	₹1,20,000
Profit	₹50,000

2) From the following data of **ABC Ltd.**, calculate:

- Total Sales = ₹3,00,000
- P/V Ratio = 40%
- Fixed Cost = ₹72,000

Find out:

1. Total Contribution
2. Profit or Loss
3. Break Even Sales (₹)
4. Margin of Safety (₹)
5. Percentage of Margin of Safety

1 Total Contribution

Formula:

$$\begin{aligned}\text{Contribution} &= \text{Sales} \times \text{P/V Ratio} \\ &= 3,00,000 \times 40\% = \boxed{\text{₹1,20,000}}\end{aligned}$$

2 Profit or Loss

Formula:

$$\begin{aligned}\text{Profit} &= \text{Contribution} - \text{Fixed Cost} \\ &= 1,20,000 - 72,000 = \boxed{\text{₹48,000 (Profit)}}\end{aligned}$$

3 Break Even Sales (₹)

Formula:

$$\begin{aligned}\text{BEP (₹)} &= \frac{\text{Fixed Cost}}{\text{P/V Ratio}} \\ &= \frac{72,000}{40\%} = \boxed{\text{₹1,80,000}}\end{aligned}$$

4 Margin of Safety (₹)

Formula:

$$\begin{aligned}\text{MoS} &= \text{Actual Sales} - \text{BEP Sales} \\ &= 3,00,000 - 1,80,000 = \boxed{\text{₹1,20,000}}\end{aligned}$$

5 Percentage of Margin of Safety

Formula:

$$\begin{aligned}\text{MoS \%} &= \frac{\text{Margin of Safety}}{\text{Actual Sales}} \times 100 \\ &= \frac{1,20,000}{3,00,000} \times 100 = \boxed{40\%}\end{aligned}$$

Particulars	Answer
Total Contribution	₹1,20,000
Profit	₹48,000
BEP (₹)	₹1,80,000
Margin of Safety	₹1,20,000
MoS %	40%